Growth Rates in Ergodicity Economics

Beyond the Log Transform

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EE 2026

EE is often reduced to “take the log” — this misses the point

What people often think EE says

Take the log of your wealth.
Maximise its expected value.

Equivalently: use log utility. The log is the whole story.

What EE actually says

Find the **time-average growth rate** of your wealth under its **actual dynamics.**
Maximise that.

The log is one answer to “what are the dynamics?” — not the only one.

What does “actual dynamics” mean — and why does it matter?

A growth rate is only meaningful when measured on the right clock

EE axiom

Maximize the **time-average growth rate**: $g = \lim_{T\to\infty} \frac{1}{T}\left[\varphi(x(T)) - \varphi(x(0))\right]$

φ is the **ergodicity transform** — the function that steadies the clock.

Additive dynamics

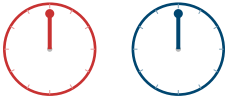
$$x(t+1) = x(t) + \xi$$

- Increment ξ is independent of x
- Natural unit of change: Δx
- Growth rate: $g = \mathbb{E}[\xi]$

Multiplicative dynamics

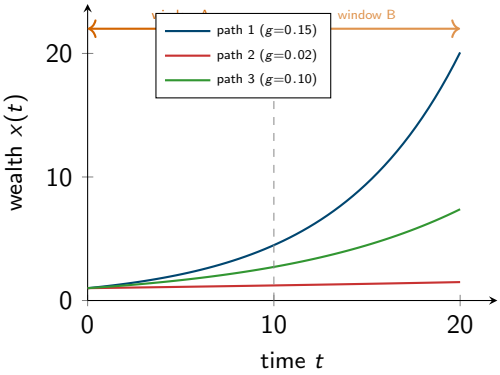
$$x(t+1) = x(t) \cdot \eta$$

- Return η is independent of x
- Natural unit of change: $\Delta \ln x$
- Growth rate: $g = \mathbb{E}[\ln \eta]$



The wrong clock gives a different answer every time

Three paths, same log growth rate



Linear clock: arithmetic return $\frac{\Delta x}{\Delta t}$

	linear clock		log clock
	Win. A	Win. B	(stable)
Path 1	0.35	1.56	0.15
Path 2	0.02	0.03	0.02
Path 3	0.17	0.47	0.10

Path 1: linear reading jumps 4× between windows.

Log reading: constant throughout.

If your clock is wrong, your growth rate is fiction.

$10 now or $25 in a month? Additive world: timing is all that matters

A stranger offers you two certain payments:

- Option 1: $10 **now**
- Option 2: $25 **in one month**

Under **additive dynamics** ($x \to x + \xi$), the growth rate is the payment divided by the waiting time:

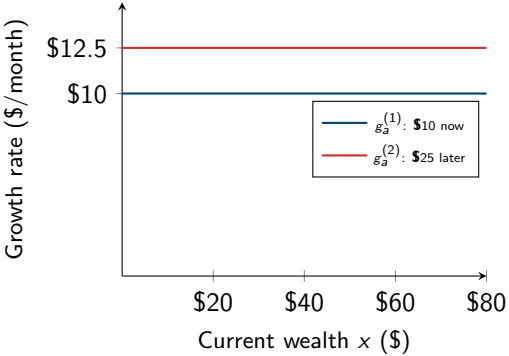
$$g_a^{(1)} = \frac{\$10}{1\,\text{month}}, \qquad g_a^{(2)} = \frac{\$25}{2\,\text{months}}$$

$g_a^{(2)} = \$12.50/\text{month} > g_a^{(1)} = \$10/\text{month}$

Always prefer $25. No dependence on wealth.

The additive clock reads only the absolute increment — your current wealth is invisible to

Growth rate does not depend on wealth



$10 now or $25 in a month? Multiplicative world: your wealth decides

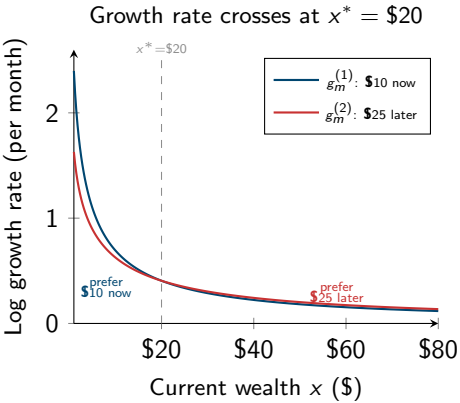
Under **multiplicative dynamics** ($x \to \eta x$), the right clock is $\ln x$.

Same additive payments — evaluated on the log clock, because that is what matches the dynamics.

Monthly log-clock growth rates:

$$g_m^{(1)} = \ln \frac{x+10}{x}, \qquad g_m^{(2)} = \frac{1}{2} \ln \frac{x+25}{x}$$

The discount rate $\delta = g$ follows directly — derived from dynamics, not assumed.



The log clock works for multiplicative dynamics — but wealth can grow in other ways

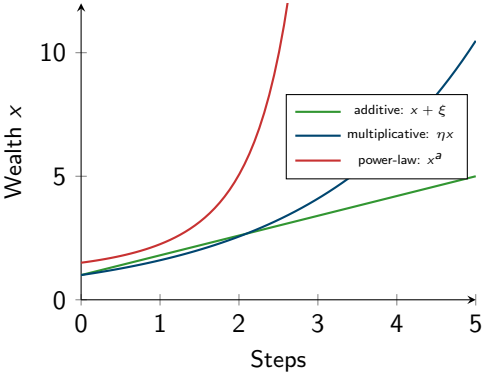
So far: $x \to \eta\, x$ — each step *multiplies* wealth.

But some processes go further:
$x \to x^a$ — each step *raises wealth to a power*.

Not x scaled by a factor,
but x taken to an exponent.

On the log clock: $\ln(x^a) = a \ln x$ — the increment *still depends on the current value*. The log clock cannot keep up.

Three qualitatively different growth curves



We need a faster clock.
What is it?

Power-law dynamics need a faster clock: $\varphi(x) = \ln\ln x$

Finding the clock for $x \to x^a$:

Let $y = \ln x$. $\quad x \to x^a \;\Rightarrow\; y \to a\,y$ — still multiplicative.

Let $z = \ln y = \ln\ln x$. $\quad z \to \ln(a\,y) = \ln a + z$ — additive!

$$\boxed{\varphi(x) = \ln\ln x}$$

Growth rate on the double-log clock:

$$g = \ln a \quad \text{(constant — no dependence on } x\text{)}$$

e.g. $x \to x^2$: $g = \ln 2$. $\quad x \to x^{1/3}$: $g = -\ln 3$.

Three dynamics — three clocks

Dynamics	Clock φ	g
$x \to x + \xi$	x	$\mathbb{E}[\xi]$
$x \to \eta\,x$	$\ln x$	$\mathbb{E}[\ln \eta]$
$x \to x^a$	$\ln\ln x$	$\mathbb{E}[\ln a]$

The clock stops at ruin — and that changes the optimal strategy

The time-average framework assumes the clock runs to infinity.

But real processes have **absorbing states**: bankruptcy, ruin, extinction.

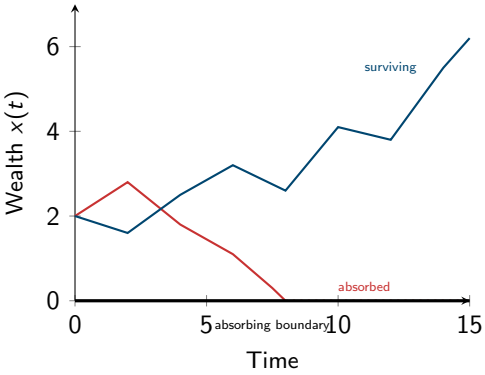
Once absorbed, the clock stops permanently — no future growth can compensate.

Consequence

Maximising g alone is not enough near a boundary.

The optimal strategy must weigh the growth rate **against** the probability that the clock stops before the long run arrives.

Same dynamics, different fates



EE says: find the right clock for your dynamics, then maximise time-average growth

Three take-home messages:

1. **EE is not the log transform**
   The log is the right clock *for multiplicative dynamics*. Different dynamics demand different clocks.

2. **Discount rates are not free parameters**
   Your clock speed equals your discount rate. It is derived from the dynamics, not fitted to data.

3. **The clock can stop**
   Near absorbing boundaries, maximising g alone is not enough — the probability of ruin changes what you should do.

The question EE asks

What is the right clock for your dynamics?

Additive	$\varphi = x$
Multiplicative	$\varphi = \ln x$
Power-law	$\varphi = \ln \ln x$
...	...